

MARCO BELLINGERI

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PROFESSIONAL PROFILE

Software engineer with cross-functional experience across development, infrastructure and application security. I have designed and shipped enterprise platforms to production — debt collection, critical-infrastructure monitoring, video streaming for Rome Metro C — with security-by-design at every level: OWASP hardening, personal-data encryption, threat modeling, zero-trust networking. Core stack: Python, React/TypeScript, AWS, Kubernetes, CI/CD with security gates. Remote-first.

TECHNICAL SKILLS

Backend & API Python (FastAPI, Flask, Falcon, Django) | REST API Design | Clean Architecture | DDD

Cloud & Infrastructure AWS (Lambda, SQS, SNS, SES, S3, EC2, IAM, VPC, CloudWatch) | Oracle Cloud OCI | Google Cloud | Cloudflare

Containers & Orchestration Docker | Kubernetes | K3s (production) | Helm | k3d | Terraform | Linux/systemd

Databases PostgreSQL + PostGIS | MySQL | Redis | MongoDB | SQLite FTS5 | Alembic migrations

DevOps & Monitoring CI/CD (GitHub Actions, Jenkins) | Prometheus | Grafana | Sentry | Cloudflare Tunnel

Frontend React | TypeScript | JavaScript ES6+ | Redux/Redux-Saga | Astro | Material-UI | PWA

AI / ML NLP (HuggingFace Transformers, Groq, Anthropic Claude API) | Sentiment Analysis | Emotion Detection | Claude Code | Model Context Protocol (MCP)

Cybersecurity & Secure Development

- **Security Automation:** CI/CD pipelines with Bandit, pip-audit, CodeQL, Gitleaks — vulnerabilities blocked before merge across 6 projects.
- **OWASP Hardening:** full Top 10 audit with documented fixes (brute-force lockout, CSRF middleware, CSP/HSTS, rate limiting, session hardening) — applied in production.
- **Applied Cryptography:** AES-256/Fernet for encryption at rest of PII, HMAC-SHA256 for API key hashing, bcrypt with timing-safe comparison.
- **Critical Infrastructure:** 24/7 supervision of 400+ video endpoints (Rome Metro Line C) with credential masking in logs and binary obfuscation. Enterprise network diagnostics (Etihad Airways) with Scapy deep path analysis.
- **Zero-trust networking:** Cloudflare Tunnel in production, K3s secrets management, presigned URLs for isolated file uploads.
- **Threat modeling:** SECURITY.md with documented attack surface across 4 public repositories, GDPR by design.

PROFESSIONAL EXPERIENCE

VETRERIA MONFERRINA — Casale Monferrato (AL), Remote | March 2026 – Present

Full-Stack & AI Agent Developer | independent project for a real client

Production website and SEO/AEO AI agent, designed and run independently for a real client.

- **Website:** Astro 6 SSG/SSR + Sanity CMS, 31 pages, Lighthouse 100/100/100 (including Accessibility WCAG 2.1 AA). Vitest 138 tests + Playwright E2E, SonarCloud Quality Gate. Security: CSP/HSTS/X-Frame-Options, rate limiting + honeypot + CSRF on the quote form, Cloudflare WAF, CodeQL + Dependabot.
- **Monferrino:** scheduled AI agent (GitHub Actions cron, zero servers) for SEO/AEO and competitor monitoring — RAG on Supabase pgvector with Voyage AI embeddings, Firecrawl scraping, reviewable micro-PRs against the site.
- **AI Act compliance:** human-oversight logging on every run, DPIA/FRIA in progress — real AI governance in production, not just theory.
- End-to-end monitoring with Checkly across both repositories.

github.com/Monferrina/vetrieriamonferrina.com | github.com/Monferrina/monferrinoAI | Live: vetrieriamonferrina.com

Stack: Astro, TypeScript, Sanity CMS, Node.js, Supabase + pgvector, Voyage AI, Firecrawl, GitHub Actions, Vercel, Cloudflare, Checkly.

FAIRFIELD S.p.A. — Figline e Incisa Valdarno (FI), Remote

Full-Stack Software Engineer | December 2025 – February 2026

Development and maintenance of a complete enterprise platform for debt collection management, comprising three integrated projects:

- **Backend API:** Clean/DDD backend on Falcon (Python 3.10): 100+ REST endpoints, 50+ MySQL tables, AWS integration (Lambda, SQS, SNS, SES, S3, Pinpoint), 35+ background processors and 23+ Lambdas for async processing.
 - **Commissions System:** Automated monthly commission calculation for collectors: Pandas/NumPy analysis, Excel/PDF/CSV/HTML reports, PyQt6 interface and CLI.
 - **Frontend SPA:** React/TypeScript SPA with Material-UI for a multi-role interface (Collector, Debtor, Back Office, Lawyers). Redux/Redux-Saga, real-time geolocation with Google Maps, push notifications (OneSignal), IT/EN i18n, PWA.
- Stack:** Python, Falcon, React, TypeScript, MySQL, Redis, AWS (Lambda, SQS, SNS, SES, S3), Google Maps API, OpenAI API, Docker, systemd.

GECO S.R.L. — Galliate (NO), Italy

Software Engineer | April 2025 – December 2025

Design and development of software solutions for the monitoring, security and automation of critical infrastructure.

- **RTSP Streaming Supervisor – Metro Line C, Rome:** Python system for the 24/7 supervision of RTSP video streams across 400+ monitors (stations and trains). Management of critical processes (MediaMTX, ffmpeg, ConVis) with automatic restart and progressive backoff. Logging with credential masking, automatic rotation and per-process separation. Graceful shutdown on SIGINT/SIGTERM. Protected distribution with Nuitka and binary obfuscation. Collaboration with a German company for the ConVis integration.
- **SmartAirway Diagnostics – Etihad Airways:** Cross-platform Python tool for advanced enterprise network diagnostics. Analysis of segments of the Etihad Airways network to identify problematic hops and nodes. ICMP/traceroute deep path analysis with Scapy, DNS checks, speedtest, network interface monitoring with psutil. Automatic CSV logging for post-mortem analysis. CI/CD pipeline with pre-commit hooks and integrated security scanning (Bandit, pip-audit).

Stack: Python, ffmpeg, MediaMTX, Scapy, Nuitka, systemd, GitHub Actions, Oracle Cloud OCI (Prometheus/Grafana).

PEROPTYX — Italy, Remote

Data Analyst | September 2024 – March 2025

Analysis, evaluation and optimization of geographic data and digital maps for navigation services. Worked with Google Maps, Excel, Python and proprietary software.

FONDAZIONE L. CLERICI — Mortara (PV), Italy

Computer Science Teacher, Upper Secondary School | September 2021 – September 2024

Taught Computer Science, Mathematics and Information Security at a vocational upper secondary school. Designed and delivered courses on programming, networking, operating systems, cryptography and data security.

FIWAY S.R.L. — Alessandria, Italy

IT Engineer | June 2021 – September 2021

Built a web-based cloud monitoring platform for IoT and Industry 4.0. Data analysis (SQL), Linux server administration, VM configuration, monitoring with Prometheus and Grafana, VPN management and network security.

PERSONAL PROJECTS

marcobellingeri.dev — Security-first personal site with a RAG-powered magazine

marcobellingeri.dev | github.com/MK023/marcobellingeri.dev

- Static bilingual Astro site on Cloudflare Workers: hash-based CSP with no unsafe-inline, Mozilla Observatory A+; the Security section re-reads its own headers live — the site demonstrates, it does not claim.
- ask terminal: RAG on Supabase pgvector (Voyage AI embeddings, Claude generation) with OWASP LLM hardening — rate limiting, Turnstile, model output treated as untrusted input, EU AI Act art. 50 disclosure.
- Editorial pipeline on autopilot with human gates: verified sourcing, LLM-as-a-judge in CI, supply-chain L3 (SBOM, attestation verified at deploy, SHA pinning).
- Radar: aggregator of official CERT bulletins (CISA, NCSC UK, CERT-FR) with a licence register enforced in CI.

Stack: Astro, TypeScript, Cloudflare Workers, Supabase pgvector, Anthropic Claude API, Voyage AI, GitHub Actions, Sentry, Langfuse.

JobSearch Command Center — AI-Powered Job Search Platform

github.com/MK023/JobSearch

- Microservices architecture: FastAPI backend + Nginx reverse proxy + PostgreSQL + Redis cache, Docker Compose, MCP server for AI agent integration.
- AI analysis with Claude Haiku/Sonnet: CV-to-job match score (0–100), gap analysis with severity levels and action plans, interview question generation.
- Feature set: multilingual cover letters, follow-up emails, LinkedIn outreach messages, recruiter CRM, Glassdoor cache, API cost tracking, document scanning (PDF/DOCX/XLSX).
- Security: full OWASP Top 10 audit, brute-force lockout (5 attempts / 30 min), CSRF Origin validation, session hardening (SameSite strict, 4h expiry), CSP/HSTS/X-Frame-Options, multi-tier rate limiting, CodeQL + Bandit + pip-audit in CI. 342 tests, mypy strict with 0 errors.
- Deployment: Fly.io with auto-sleep, CI/CD on GitHub Actions (8 stages: lint, format, type-check, security audit, test, Docker build, CodeQL, deploy).

Stack: Python, FastAPI, PostgreSQL, Redis, Nginx, Anthropic Claude API, Docker Compose, Fly.io, GitHub Actions.

HappyKube — AI-Powered Emotion Analysis Telegram Bot (2+ years)

github.com/MK023/HappyKube

- Multilingual emotion detection (7 emotions) via Groq LLaMA 3.3 70B with integrated sentiment analysis.
- Production-grade architecture: Clean Architecture and DDD. RESTful Flask API with PostgreSQL, Redis caching, Alembic migrations.
- Security hardening: Fernet encryption for PII at rest, prompt injection prevention, cache poisoning prevention, API key authentication with hashing, webhook secret validation, input sanitization pipeline.

Stack: Python, Flask, PostgreSQL, Redis, HuggingFace Transformers, Groq, Docker, Render.com.

MD Vault — Self-Hosted Knowledge Base with K3s and Terraform

github.com/MK023/md_vault | Live: mdvault.site

- K3s/Kubernetes orchestration: local deployment via k3d, native K3s on GCE e2-small (production); complete manifests.
- Infrastructure as Code: Terraform for Google Cloud provisioning (VPC, firewall, compute, GCS backend state), Cloudflare Tunnel.
- FastAPI backend with SQLite FTS5 (BM25 full-text search), multi-format upload, built-in viewers. Win95 frontend (vanilla JS/CSS, zero build step).
- Security: zero-trust access via Cloudflare Tunnel, XSS prevention (DOMPurify), path traversal protection, JWT + bcrypt authentication, K3s secret management, backup CronJob to Cloudflare R2.

Stack: Python, FastAPI, SQLite FTS5, K3s, Kubernetes, Terraform, Google Cloud, Cloudflare Tunnel, Docker, Sentry.

RabbitWatch — Super Monitor & Control Plane Stack

github.com/MK023/RabbitWatch

- Monitoring and self-healing stack for cloud/on-premise infrastructure. FastAPI for active health checks on critical services (VPN, NAS, RabbitMQ, databases, VMs) with automatic recovery through a centralized Control Plane.
- Full monitoring: Prometheus + Alertmanager + Grafana for metrics and alerting, RabbitMQ for an event-driven architecture, Python producers/consumers for metric collection and recovery handling.
- Infrastructure: designed for AWS EC2 with IAM and VPC for isolation and security, horizontal and vertical scalability. Docker Compose orchestration, Ubuntu 22.04 LTS, systemd integration.

Stack: Python, FastAPI, Prometheus, Grafana, Alertmanager, RabbitMQ, MongoDB, Docker, AWS (EC2, IAM, VPC), Ubuntu 22.04 LTS.

TorinoParking — Real-Time Parking Availability API — Turin

github.com/MK023/TorinoParking

- Async FastAPI (Python 3.12) with PostgreSQL 16 + PostGIS for geospatial queries, Redis 7 for caching with orjson/zlib compression and conditional requests via ETag (304).
- Security-first API design: HMAC-SHA256 API key hashing (zero plaintext storage), multi-tier sliding-window rate limiting (20/100/1000 req/min), security headers, Pydantic input validation, threat model documented in SECURITY.md.
- 51 tests (32 unit + 18 integration + 1 e2e) with PostgreSQL/Redis testcontainers; CI/CD on GitHub Actions; structured logging with structlog.

Stack: Python, FastAPI, PostgreSQL + PostGIS, Redis, APScheduler, Docker, structlog, Sentry, GitHub Actions.

EDUCATION

Università degli Studi del Piemonte Orientale — Vercelli, Italy

BSc in Computer Science | September 2016 – in progress

Polish-Japanese Academy of Information Technology (PJAIT) — Warsaw, Poland

Computer Science (Erasmus programme) | July 2018 – March 2019

Istituto Tecnico Commerciale Leardi — Casale Monferrato (AL), Italy

Accounting & Business Administration Diploma – I.G.E.A. | August 2005 – July 2010

CERTIFICATIONS

- IT Specialist – Cybersecurity — Cisco Networking Academy | 2022
- AWS Cloud Practitioner + Cloud Computing — Tree Opinio | 2022
- Linux Essentials — Linux Professional Institute | 2022
- GitHub Foundations — GitHub | 2025
- MongoDB Python Developer Path — MongoDB | 2024
- Python PCEP [30-02] — Python Institute | 2024
- Foundational C# with Microsoft — freeCodeCamp | 2025
- Containers & Kubernetes Essentials — IBM | 2024
- Docker Essentials: A Developer Introduction — IBM | 2024
- Anthropic Academy — 15 courses completed, including: Claude Code in Action, Building with the Claude API, Introduction to Model Context Protocol, Introduction to Agent Skills, Introduction to Subagents, AI Fluency: Framework & Foundations | 2026

LANGUAGES

- Italian: native speaker.
- English: B2 (listening, reading, spoken production, spoken interaction, writing).

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